

KLH UNIVERSITY

Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-26)
Faculty Name	Ms Md. Razia
Student Name	Adunuri Bhavana sri
Roll Number	2520030232
Branch	CSE
Course Outcome	5

Case Study Topic: HealthNet – Intelligent Hospital Resource Management System using Advanced Quick Sort

Work Done Summary:

- Implemented Dijkstra's Shortest Path algorithm in Java to optimize the emergency transportation latency and clinical logistics routing paths between regional medical centers, critical supply reserves, and receiving hospitals.
- Modeled medical centers, ambulance hubs, and central pharmacy deposits as explicit nodes within a weighted undirected network topology graph.
- Utilized dynamic edge weights representing traffic density, physical distance, and emergency travel time overhead parameters.
- Supported instant shortest-path computations to safely direct critical life-support resources toward emergency crisis points without routing starvation.
- Compared single-source shortest path execution correctness across distinct graph weights to ensure system performance guarantees.

Implementation Details:

- **1. Configured PatientRanker system** utilizing contiguous array structures to manipulate multiple patient records simultaneously.
- **2. quickSortPartition()** applies a low-overhead pivot layout to slice data fields into high-severity and low-severity sub-arrays.

- **3. executeRecursiveSort()** triggers log-linear tree division logic to sort the entire sub-array structure sequentially.
- **4. In-place data swap function** moves target score cells using basic pointers without creating redundant data duplication objects.
- **5. Main execution routine** loads unstructured trauma center severity score distributions, tracks algorithm runtimes, and prints sorted triage registers.

Result: Screenshots/Output

```
PS D:\JavaPrograms> javac HealthNetQuickSort.java
PS D:\JavaPrograms> java HealthNetQuickSort

HealthNet - Advanced Triage Prioritization & Sorting System (C05)
Loading Raw Unsorted Emergency Patient Severity Index Rows:
Unsorted Array: [67, 92, 45, 89, 71, 99, 54, 80]

Initiating Quick Sort Strategy utilizing Randomized Pivot Allocation Rules...
Slicing sub-arrays and updating memory bounds sequentially...

Sorted Triage Ranking Register (Descending Priority Order):
Priority 1: Severity Score 99 (Patient Ref: P-106)
Priority 2: Severity Score 92 (Patient Ref: P-102)
Priority 3: Severity Score 89 (Patient Ref: P-104)
Priority 4: Severity Score 80 (Patient Ref: P-108)
Priority 5: Severity Score 71 (Patient Ref: P-105)
Priority 6: Severity Score 67 (Patient Ref: P-101)

Empirical Benchmarking Report: Sorting achieved successfully. 0 thread blocks
identified.
```

Time Complexity & Empirical Analysis:

- Quick Sort Time Complexity: Average case $O(n \log n)$; Worst-case $O(n^2)$ bypassed via randomized pivot constraints.
- Auxiliary Memory Allocation: $O(\log n)$ call stack space during recursive tree subdivisions.
- Quick Sort provides exceptionally high cache efficiency and minimal auxiliary space requirements, making it ideal for intensive tracking modules.

GitHub Repository Link:

<https://github.com/bhavana-sri25/Sketch-DSA-C05>

Student Signature

Faculty Signature